CEGFNet: Common Extraction and Gate Fusion Network for Scene Parsing of Remote Sensing Images

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***Abstract*— Scene parsing of high spatial resolution (HSR) remote sensing images has achieved notable progress in recent years by the adoption of convolutional neural networks. However, for scene parsing of multimodal remote sensing images, effectively integrating complementary information remains challenging. For instance, the decrease in feature map resolution through a neural network causes loss of spatial information, likely leading to blurred object boundaries and misclassification of small objects. In addition, object scales on a remote sensing image vary substantially, undermining the parsing performance. To solve these problems, we propose an end-to-end common extraction and gate fusion network (CEGFNet) to capture both high-level semantic features and low-level spatial details for scene parsing of remote sensing images. Specifically, we introduce a gate fusion module to extract complementary features from spectral data and digital surface model data. A gate mechanism removes redundant features in the data stream and extracts complementary features that improve multimodal feature fusion. In addition, a global context module and a multilayer aggregation decoder handle scale variations between objects and the loss of spatial details due to downsampling, respectively. The proposed CEGFNet was quantitatively evaluated on benchmark scene parsing datasets containing HSR remote sensing images, and it achieved state-of-the-art performance.**

***Index Terms*— Gate fusion module (GFM), high spatial resolution (HSR) remote sensing image, multimodal fusion, scene parsing.**

I. Introduction

HIGH spatial resolution (HSR) remote sensing images can capture various geospatial objects, including airplanes, ships, vehicles, and buildings. Identifying such objects on HSR remote sensing images has a great utility for urban monitoring and management [1]. Aerial image parsing is an important task in remote sensing because it can provide semantic and localization information cues for targets of interest. In scene parsing, a semantic category is assigned to every image pixel.

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For high-resolution image processing, scene parsing and semantic labeling are particularly beneficial. Recently, convolutional neural networks (CNNs) have become predominant in visual recognition, achieving state-of-the-art (SOTA) performance on a variety of problems, including scene parsing [2]–[4]. However, CNN-based scene parsing of HSR remote sensing images faces various challenges, including the high diversity of geospatial objects and the class imbalance originating from the high resolution [5]. High diversity leads to high intraclass heterogeneity and low interclass variance. For example, low vegetation and trees can be confused, as well as objects and humans in urban areas [6]. Class imbalance refers to the different proportions of objects in an image and is characterized by uneven distributions across categories.

In addition to the abovementioned problems, it is difficult to distinguish regions with similar spectral characteristics using only multispectral data in HSR remote sensing images. Recent studies have shown that the introduction of information from multiple sources can effectively promote the robustness of scene parsing. For instance, digital surface models (DSMs) have been utilized to extract supplementary features that further improve scene parsing [7], [8]. A DSM provides elevation data of objects in a remote sensing image, facilitating the segmentation of tall objects, such as buildings and trees. However, the inherent differences between DSMs and multispectral data hinder the proper extraction and fusion of complementary information between these two data modalities through operations, such as elementwise addition [7]. In fact, more noise may be introduced, undermining the parsing performance.

To solve these challenging problems, we propose a deeply supervised common extraction and gate fusion network (CEGFNet) that integrates features from two information modalities and combines high-level semantic features, low-level spatial details, and global context information to segment complete objects in HSR remote sensing images.

The main contributions of this study are summarized as follows.

1) The novel end-to-end CEGFNet for scene parsing of multimodal HSR remote sensing images is proposed. CEGFNet uses a gate mechanism to filter out redundant information from multimodal data and uses their similarities to extract complementary information for feature fusion. It also handles the high variability of object

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scales often encountered in remote sensing images and prevents the loss of spatial details due to pooling, which enlarges the receptive field in conventional scene parsing.
2) A gate fusion module (GFM), a global context module (GCM), and a multilayer aggregation decoder (MAD) are easily inserted into an encoder–decoder architecture for achieving high parsing accuracy.
3) The proposed CEGFNet achieves SOTA performance as demonstrated through various experiments, especially for small objects (e.g., objects in the scale of cars) on two benchmark scene parsing datasets.

The remainder of this article is organized as follows. Section II discusses the related work. Then, we detail the proposed CEGFNet in Section III. Section IV reports and analyzes comprehensive experimental results. In Section V, we present concluding remarks.

II. Related Work

We briefly discuss various existing RGB and multimodal scene parsing methods, especially in terms of their advantages and fusion strategies.

A. RGB Scene Parsing

Early scene parsing models rely heavily on handcrafted features and use random forests and various enhancements to predict class probabilities. In addition, probabilistic models called conditional random fields are often applied to refine the parsing results [9], [10]. In recent years, with the emergence of large datasets and high-performance graphics processing units (GPUs) and the development of CNNs, considerable progress has been achieved in scene parsing. Long *et al.* [4] improved the effectiveness of scene parsing by adapting a classification network to a complete convolutional network. Specifically, they concatenated the feature map of each convolutional layer of U-Net [11] to the corresponding upsampling layer such that the feature map of each layer can be utilized in subsequent calculations. U-Net has shown high performance for medical image segmentation. Badrinarayanan *et al.* [12] adopted an encoder–decoder framework, with the decoder being used to map a low-resolution encoded feature onto a full-resolution feature map. The decoder uses a pooled index of the max-pooling layer to perform nonlinear upsampling, which can improve the edge characterization and reduce the number of training parameters. This upsampling structure can be used in any encoder–decoder network. Chen *et al.* [13] adopted atrous spatial pyramid pooling (ASPP) to explore multiscale convolution features. The global background is encoded according to the image hierarchy to obtain the features, achieving SOTA performance. To reduce the consumption of memory and computing resources and increase the receptive field, these methods have been mostly based on encoders, which gradually reduce the feature map resolution and capture high-level semantic information, and decoders, which recover spatial information.

Hu *et al.* [14] adopted an attention mechanism and proposed a squeeze-and-excitation network that mainly learns the correlation between feature channels and screens out the attention for channels, thus slightly increasing the computational burden but providing high parsing performance. Woo *et al.* [15] proposed a convolutional attention module, which achieved high performance by integrating a spatial attention block and a channel attention block successively to emphasize meaningful features along the spatial and channel dimensions. In contrast, Fu *et al.* [16] integrated spatial attention and channel attention in parallel and fused the output of the two attention mechanisms, further enhancing feature representation. Hou *et al.* [17] reformulated spatial pooling by introducing a method called strip pooling, which considers a long narrow kernel (i.e., $N \times 1$ or $1 \times N$). Based on strip pooling, SPNet was developed with a spatial pooling architecture, and its effectiveness was demonstrated through several experiments. Zhang *et al.* [18] proposed a residual atrous pyramid network to sequentially aggregate global-to-local context cues through residual refinement.

Despite achieving high performance, it is difficult to distinguish regions with similar spectral features using only multispectral data or RGB data in scene parsing of HSR remote sensing images. Thus, multimodal information has been recently studied aiming to improve the robustness of the corresponding parsing methods.

B. Multimodal Scene Parsing

Multimodal data fusion has gained considerable attention for exploiting the complementarity in data from different sources [19]–[22]. Early single-modal scene parsing methods mainly consider the combination of low-level spatial details with high-level semantic information. In contrast, deep multimodal models further incorporate intermediate fusion or hierarchical fusion because CNNs can learn representative features from data across different levels. In addition, early fusion can intuitively reuse conventional single-modal scene parsing methods.

1) RGB-D Scene Parsing: Successful RGB-D (RGB and depth) scene parsing depends on effectively integrating multimodal features. Ding *et al.* [23] adapted a multiscale RGB network for RGB-D scene parsing by concatenating the input depth and RGB streams. Wang *et al.* [24] aimed to aggregate the high-level semantic information of the RGB and depth modalities using independent networks. Zhou *et al.* [22] and Chen *et al.* [25] applied the HHA representation to depth data, encoding the horizontal disparity, height above ground, and angle with gravity per pixel. Hazirbas *et al.* [26] proposed FuseNet for the bottom-up fusion of multilevel depth features into an RGB encoder. Jiang *et al.* [27] extended FuseNet by fusing multilevel features along a top-down path. Similarly, Lee *et al.* [28] proposed a multimodal feature module and a multilevel feature refinement module to fuse multilevel features along a top-down path. Hu *et al.* [29] proposed an attention complementary module and architecture with three parallel branches, achieving SOTA performance on RGB-D indoor datasets. Lin *et al.* [30] proposed a switchable context module to facilitate RGB-D scene parsing. Depth data were used to identify objects in multiple regions. Zhou *et al.* [31] proposed a three-branch self-attention

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module comprising two asymmetric input branches and a cross-modal distillation branch with a self-attention block to perform cross-modal feature fusion. Zhou *et al.* [32] proposed a coattention module to capture fine-grained interactions between RGB and depth features. Zhang *et al.* [33] proposed a nonlocal aggregation module with a multimodal nonlocal aggregation module to exploit the nonlocal context of RGB-D features at multiple stages, thus achieving high performance in RGB-D scene parsing. Chitta *et al.* [34] proposed quadtree generating networks (QGNs) for RGB-D scene parsing. Seichter *et al.* [35] proposed an efficient RGB-D scene parsing method, ESANet, characterized by two enhanced ResNet-based encoders. Yue *et al.* [36] proposed a two-stage cascaded decoder for RGB-D scene parsing.

2) Multimodal Scene Parsing Based on Multispectral Data and DSMs: Given the difficulty to obtain multispectral data and DSMs, few studies are available on scene parsing based on these data sources. Peng *et al.* [7] proposed an architecture that combines dense connections and fully convolutional networks, called DFCN, to automatically obtain fine-grained scene parsing results. They also investigated a multimodal network that incorporated DSMs into DFCN. Although DSMs provide the elevation data of objects, they introduce more noise than depth data. Similar to RGB-D scene parsing, the effective extraction and integration of complementary information between multimodal data are also challenging for multispectral data and DSMs. Peng *et al.* [7] simply fused features from the two modalities through elementwise addition, failing to meaningfully fuse data.

Compared with RGB and RGB-D scene parsing, parsing multimodal HSR remote sensing images is more challenging due to the noisy DSM data. We aimed to find correlations between the two types of data after filtering by a gate mechanism to extract complementary multimodal features and improve the performance of multimodal scene parsing.

III. Proposed CEGFNet

A. Overview

The CEGFNet architecture is shown in Fig. 1 and corresponds to a deeply supervised encoder–decoder network. It integrates features between two different modalities and combines high-level semantic features, low-level spatial details, and global context information to accurately identify complete objects in HSR remote sensing images. ResNet-50 [37] is the CEGFNet backbone. To fit the scene parsing task, we remove all the fully connected layers and the last max-pooling layer of ResNet-50. Fed with two data modalities, each of the two ResNet-50 networks uses five convolutional blocks (Conv1, Res2, Res3, Res4, and Res5) to extract features at multiple scales. CEGFNet also incorporates several GFMs into a two-stream backbone network to extract complementary features between the two modalities. Then, the GCM expands the receptive field through dilated convolutions with different dilation rates to capture multiscale features, and an attention mechanism reduces feature redundancy. Although pooling enlarges the receptive field in the bottom-up feature encoder,

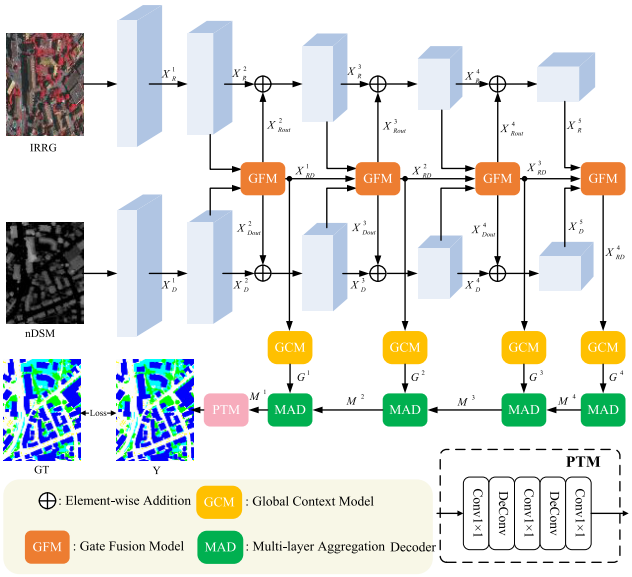


Fig. 1. Architecture of the proposed CEGFNet (PTM: prediction terminal module).

it leads to the loss of many spatial details. Therefore, we apply the MAD to use multiscale information from the GCM. As a result, during top-down decoding, multiscale features are gradually integrated and refined.

We detail the GFM in Section III-B. Then, we introduce the GCM in Section III-C to mitigate the effects of large-scale variations in objects captured in remote sensing images. Thereafter, we detail the MAD in Section III-D to improve the spatial details in a top-down process. The MAD replaces the pooling that can increase the receptive field but causes spatial detail loss. Additional details about CEGFNet and its training are also provided in the following.

B. GFM

For scene parsing of multimodal HSR remote sensing images, it is essential to determine the proper use of IRRG (infrared–red–green) and normalized DSM (nDSM) information. The inherent differences between these two data modalities may lead to increasing noise or redundancy when combining features directly through operations, such as summation or concatenation. To effectively capture the complementary features of these modalities, we propose the GFM that progressively extracts joint features for proper feature fusion.

In CEGFNet, the first to fourth GFMs have similar main structures. The second and third GFMs have three inputs and three outputs, as shown in Fig. 2(a). The first GFM has two inputs and three outputs such that the pipeline remains unchanged [see Fig. 2(a)], while only the intermediate input is removed, as shown in Fig. 2(b). The fourth GFM has three inputs and one output such that the pipeline remains unchanged [see Fig. 2(a)], while two outputs are removed, as shown in Fig. 2(c).

In the IRRG encoding backbone, the output of the ith convolutional block is denoted as X_R^i ($i = 1, 2, 3, 4, 5$). Similarly, the output of the corresponding convolutional block

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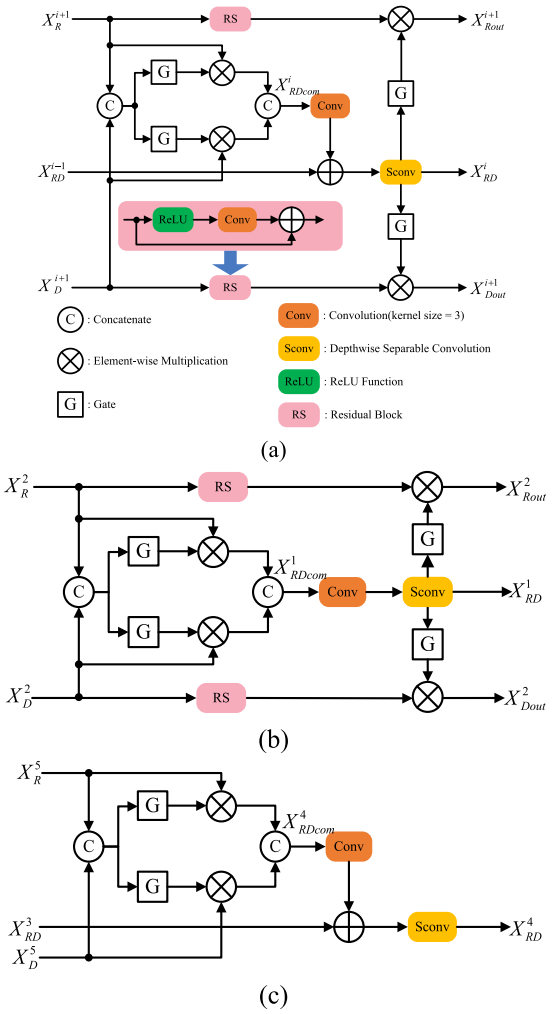


Fig. 2. GFM architecture. (a) Second and third GFMs. (b) First GFM. (c) Fourth GFM.

in the nDSM encoding backbone is denoted as X_D^i ($i = 1, 2, 3, 4, 5$). The GFM architectures are shown in Fig. 2. Although X_R^1 and X_D^1 contain abundant spatial information, we remove them because they are considerably noisy. For GFM i ($= 1, 2, 3, 4$), the input features from the IRRG encoding backbone are X_R^{i+1} ($i = 1, 2, 3, 4$), and the input features from the nDSM encoding backbone are X_D^{i+1} ($i = 1, 2, 3, 4$). Then, the output features of GFM i are X_{Rout}^{i+1} and X_{Dout}^{i+1} for $i = 1, 2, 3$ and cross-modal features X_{RD}^i for $i = 1, 2, 3, 4$. Apart from the first GFM [see Fig. 2(b)], the other GFMs [see Fig. 2(a) and (c)] have input feature X_{RD} from the previous GFM. A gate is denoted as Gate and consists of two convolutional layers with a rectified linear unit (ReLU) layer in between, and a sigmoid function σ bounds the values to [0, 1]. We also adopt two residual blocks, RS, in this module. An RS consists of a ReLU layer, a convolutional layer with a 3×3 kernel, a batch normalization layer, and elementwise addition. Gate and RS are respectively formulated as

$$\text{Gate}(x) = \sigma(\text{Conv1} \times 1(\text{relu}(\text{Conv1} \times 1(x))) \quad (1)$$

$$\text{RS}(x) = \text{BN}(\text{Conv3} \times 3(\text{relu}(x))) + x \quad (2)$$

where σ denotes the sigmoid function, Conv $a \times a$ denotes a convolutional layer with an $a \times a$ kernel, relu represents the ReLU activation function, BN denotes batch normalization, and $+$ denotes the elementwise addition.

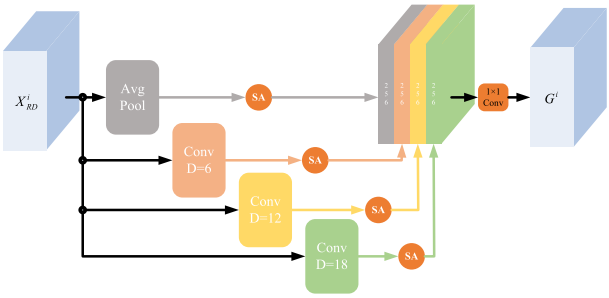


Fig. 3. GCM architecture. (D: dilation rate; Avg: average; Conv: convolution; Pool: pooling; and SA: sigmoid activation).

First, we concatenate X_R^i and X_D^i and then apply two gate units followed by multiplication by the original features to suppress redundant information. The features after multiplication are fused to obtain preliminary cross-modal features denoted as X_{RDcom}^i ($i = 1, 2, 3, 4$). After applying a convolution with a 3×3 kernel, X_{RDcom}^i is added to the cross-modal feature X_{RD}^{i-1} from the previous GFM, mitigating the loss of spatial details due to downsampling. Then, cross-modal information X of the GFM output at each layer can be obtained by applying a depthwise separable convolution. To extract similarities between cross-modal features and single-modal features, we apply elementwise multiplications to then obtain single-modal features X_{Rout}^{i+1} and X_{Dout}^{i+1} after feature refinement. We do not apply refinement to the last GFM [see Fig. 2(c)] for the single-modal features. The operations described above can be expressed as

$$X_{\text{RDcom}}^i = \text{Cat}\big(X_R^{i+1} \times \text{Gate}\big(\text{Cat}\big(X_R^{i+1}, X_D^{i+1}\big)\big), X_D^{i+1} \times \text{Gate}\big(\text{Cat}\big(X_R^{i+1}, X_D^{i+1}\big)\big)\big) \quad (3)$$

$$X_{\text{RD}}^i = \text{Sconv}\big(\text{Conv3} \times 3\big(X_{\text{RDcom}}^i\big) + X_{\text{RD}}^{i-1}\big) \quad (4)$$

$$X_{\text{Rout}}^{i+1} = \text{RS}\big(X_R^{i+1}\big) \times \text{Gate}\big(X_{\text{RD}}^i\big) \quad (5)$$

$$X_{\text{Dout}}^{i+1} = \text{RS}\big(X_D^{i+1}\big) \times \text{Gate}\big(X_{\text{RD}}^i\big) \quad (6)$$

where Cat represents concatenation, $\times$ represents the elementwise multiplication, and Sconv represents the depthwise separable convolution.

C. GCM

In HSR remote sensing images, the large-scale variations in captured objects can undermine scene parsing. Inspired by ASPP [13], we propose the GCM module [see Fig. 3]. ASPP samples an input in parallel by void convolution with different void rates and expands the feature receptive field without reducing feature spatial resolution. Although ASPP is effective for large-scale variations in objects, applying it after feature sampling may cause feature redundancy, which can be reduced by applying an attention mechanism. To reduce feature redundancy, we propose the attention-mechanism-based GCM.

The proposed GCM receives output features X_{RD}^i ($i = 1, 2, 3, 4$) from the GFM. A GCM layer consists of global average pooling and three dilated convolutions with a 3×3 kernel and dilation rates of 6, 12, and 18. In addition, we apply

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spatial attention after each layer. The output of the jth layer of the dilated convolution or pooling is denoted as X_G^j ($j = 1, 2, 3, 4$). We take the maximum and average of the features across the channel dimensions to obtain two informative feature maps, F_{max}^j and F_{mean}^j, respectively

$$F_{\text{mean}}^j = \frac{1}{C}\sum_{z=1}^{C} X_G^j(z) \tag{7}$$

$$F_{\text{max}}^j = \max_{z=1,\ldots,C} X_G^j(z) \tag{8}$$

where C denotes the number of feature channels, z represents the channel index, and j is the number of layers in the GCM.

Then, we concatenate the features along the channel dimension and use a convolutional layer to obtain a one-channel feature representation. To obtain a relatively large receptive field, a 7×7 kernel is applied for the convolution. The result is fed to a sigmoid function to obtain a normalized coordination map in [0, 1]. Then, elementwise multiplication is used between W_{sa}^j and the original X_G^j to emphasize global characteristics. The output of the jth layer of the GCM is denoted as F_G^j ($j = 1, 2, 3, 4$). We concatenate all the outputs along the channel dimension, and to reduce the computational burden, we use a convolutional layer to generate 256 channels with a 1×1 kernel. The output of the ith GCM is denoted as $G^i (i = 1, 2, 3, 4)$. The corresponding GCM operations can be expressed as

$$W_{\text{sa}}^j = \sigma\left(\text{Conv7} \times 7\left(\text{Cat}\left(F_{\text{mean}}^j, F_{\text{max}}^j\right)\right)\right) \tag{9}$$

$$F_{\text{sa}}^j = W_{\text{sa}}^j \times X_G^j \tag{10}$$

$$G^i = \text{Conv1} \times 1\left(\text{Cat}\left(F_G^1, F_G^2, F_G^3, F_G^4\right)\right). \tag{11}$$

D. MAD

To improve representative information in a top-down process, we apply the MAD [see Fig. 4] to prevent the loss of details due to pooling, which increases the receptive field. The MAD receives features G^i ($i = 1, 2, 3, 4$) from the GCM and refines them by calculating the similarity between any two locations to establish skip connections, thus refining detailed features. After receiving the features from the previous MAD improvement, the similarity between features is extracted to gradually refine and improve the features. The first three MADs have self-refining and hierarchical fusion units, and the last MAD only contains the self-refining unit. The self-refining unit is an attention module that encodes the context information into local features [16]. In this unit, the features are processed by three convolutional layers with a 1×1 kernel to reduce the number of channels and computational burden. The two features are reshaped along the H and W dimensions, and matrix multiplication is performed to calculate the similarity between the two matrices. Then, the features are normalized by softmax activation and multiplied by the initial features. In hierarchical fusion, the similarities between the MAD-processed features of the upper layer and those of the self-refining unit are obtained through multiplication, and they are fused through addition.

Then, we apply a prediction terminal module to obtain the final feature map. As shown in Fig. 1, the prediction terminal module consists of three convolutional layers with a 3×3 kernel. Between each pair of convolutional layers, a deconvolution layer expands the feature map to twice its original size. We use the common cross-entropy loss function to supervise the final output feature map

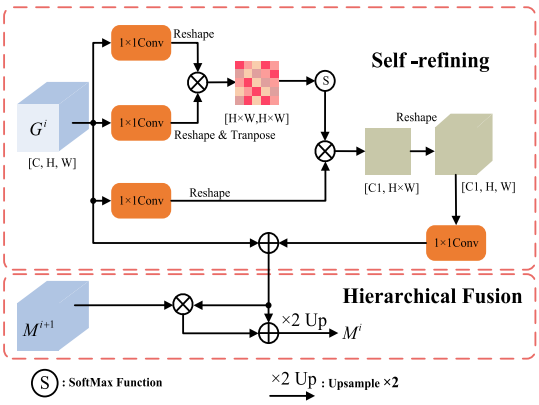


Fig. 4. MAD architecture.

$$L = -\sum_{i}^{M} Y_i \times \log(P_i), \quad i = 0, 1, 2, 3, 4, 5 \tag{12}$$

where L represents the cross-entropy loss function, M represents the number of classes, Y_i is an indicator variable being 1 if the predicted class is the same as the sample class and 0 otherwise, and P_i represents the predicted probability that the observed sample belongs to class i.

IV. Experiments and Results

A. Datasets

To evaluate the proposed CEGFNet, we used two benchmark datasets for scene parsing of HSR remote sensing images, Vaihingen and Potsdam [38]. To construct these datasets, a DSM generated from high-resolution orthogonal photographs and the corresponding dense image-matching technique were used. Both datasets cover urban scenes. Vaihingen is a relatively small village with a small multistory and many free-standing buildings, representing a typical historical city with large buildings, narrow streets, and dense settlements. Each image from the dataset was manually labeled with the six most common land cover categories, namely, impervious surfaces, buildings, trees, low vegetation, cars, and clutter/background. The clutter/background includes water bodies and objects that differ from other defined categories (e.g., containers, swimming pools, and tennis courts), which are often considered as semantic objects of no interest in urban scenes.

Vaihingen contains 33 remote sensing images of different sizes, each extracted from a larger top-level orthophotograph. The spatial resolution of both the top-level image and DSM is 9 cm. The images are formatted as 8-bit TIFF files composed of near-infrared, green, and red bands. The DSM is formatted as a single-band TIFF file with grayscale level (corresponding to the DSM height) encoded into 32-bit floating-point values. Sixteen tiles in the images were fully annotated with the six most common land cover classes. Thus, we selected 11 tiles

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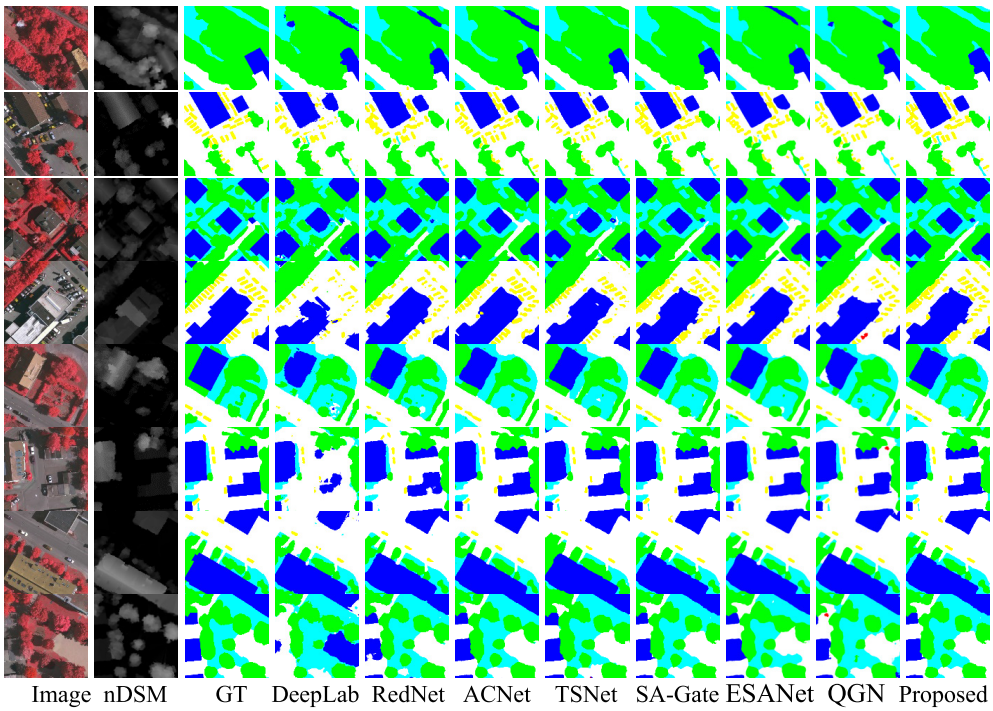


Fig. 5. Scene parsing maps obtained from CEGFNet and SOTA methods (white: impervious surfaces; cyan: low vegetation; blue: buildings; yellow: cars; green: trees; and red: clutter/background).

(1, 3, 5, 7, 13, 17, 21, 23, 26, 32, and 37) from the annotated images for the training set, and the remaining five tiles (11, 15, 28, 30, and 34) for the test set.

Similar to Vaihingen, Potsdam contains three-band remote sensing image TIFF files and a single-band DSM. The Potsdam dataset consists of 38 high-resolution aerial images with a spatial resolution of 5 cm. Twenty-four tiles of the available data were fully annotated with the same classes as for the Vaihingen dataset. We used 17 tiles (2_10, 3_10, 3_11, 3_12, 4_11, 4_12, 5_10, 5_12, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_9, 7_11, and 7_12) for the training set and seven tiles (2_11, 2_12, 4_10, 5_11, 6_7, 7_8, and 7_10) for the test set.

B. Evaluation Criteria

Unlike the evaluation criteria used in a similar study [6], we used the class accuracy (Acc), mean class accuracy (mAcc), intersection over union (IoU), and mean IoU (mIoU) [39], [40] to evaluate the parsing performance of various methods, noting that the mIoU is the conventional measure of scene parsing accuracy.

C. Implementation Details

Only 16 and 24 images were fully labeled in Vaihingen and Potsdam, respectively. The average image resolutions in Vaihingen and Potsdam are 2493 $\times$ 2063 and 6000 $\times$ 6000 pixels, respectively. Therefore, the training data are scarce, whereas the resolution of each image is very large. Thus, we cropped the original images to increase the number of training samples. As in [8], we adopted a sliding window and random cropping for the two datasets to expand the training set. Considering the resolutions of the two datasets, the input size was set to 256 $\times$ 256 pixels. In addition, we augmented the training set by applying horizontal and vertical flipping and 90°, 180°, and 270° rotations.

As in [6], we only predicted five classes in Vaihingen, omitting the clutter/background class due to the lack of training datasets for that class. On the other hand, we predicted all six classes in Potsdam. We used the PyTorch framework for the evaluations performed on a computer equipped with an NVIDIA GTX TITAN X GPU with 12 GB of memory. In the training phase, the ResNet-50 [37] blocks were initialized with pretrained parameters from the ImageNet dataset for image classification, and all the other parts were initialized randomly. The proposed CEGFNet was then trained using Adam optimization and an initial learning rate of 1e$-$4, which was subsequently divided by 10 once every 20 epochs.

D. Comparisons With SOTA Methods

We compared CEGFNet with ten SOTA deep-learning-based methods: FCN [4], SegNet [12], U-Net [11], DeepLab V3+ [13], RedNet [27], TSNet [31], SA-Gate [25], ACNet [29], QGN [34], and ESANet [35]. As the methods in [4], [11], [12], and [13] are single-modal scene parsing methods, we only used IRRG data. As the methods in [25], [27], [29], [31], [34], and [35] are intended for RGB-D scene parsing, we added the nDSM data as an additional channel replacing the depth data. The scene parsing maps for all the methods were generated by running the available codes.

1) Quantitative Evaluation: Tables I and II list the obtained Acc, IoU, mAcc, and MIoU for the proposed CEGFNet and comparison SOTA methods on Vaihingen and Potsdam, respectively. CEGFNet outperforms the comparison methods on most indicators for both datasets. Especially for cars, the scene parsing provided by CEGFNet is much better than that obtained using the other methods. In fact,

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TABLE I
EXPERIMENTAL RESULTS ON VAIHINGEN TEST SET

	Imp.surf		Building		Low veg.		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
U-Net	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
SegNet	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
DeepLabv3+	90.06	81.11	87.04	82.7	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
RedNet	91.49	84.62	94.81	91.07	78.67	66.59	**91.41**	78.27	59.77	56.06	83.23	75.32
ACNet	91.95	85.34	**95.45**	**91.82**	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
SA-Gate	90.99	**85.70**	93.85	91.72	**84.95**	**68.68**	89.06	**79.15**	**84.27**	**78.07**	**88.62**	**80.67**
TSNet	87.93	78.98	**95.81**	91.47	71.62	57.03	**94.26**	81.26	67.63	66.86	83.54	75.12
QGN	**92.46**	82.41	92.04	88.77	78.46	66.70	90.63	78.06	73.50	68.73	79.79	64.11
ESANet	**92.09**	85.18	94.93	91.16	75.72	65.48	92.35	77.65	75.92	70.11	86.20	77.92
Proposed	91.56	**85.65**	95.30	**92.15**	**82.96**	**68.62**	89.64	**79.51**	**86.89**	**78.40**	**89.27**	**80.86**

TABLE II
EXPERIMENTAL RESULTS ON POTSDAM TEST SET

	Imp.surf		Building		Low veg.		Tree		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
U-Net	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	**93.89**	90.24	50.30	36.26	82.13	72.22
SegNet	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
DeepLabv3+	**91.57**	82.49	91.78	87.59	87.36	**73.63**	85.45	73.32	**93.89**	90.04	53.80	**43.54**	83.97	75.10
RedNet	**92.19**	**82.83**	93.61	**90.13**	87.00	73.22	83.00	71.77	93.36	90.08	56.74	**43.51**	84.32	**75.26**
ACNet	91.32	82.74	**93.83**	**90.06**	86.16	73.53	**86.03**	72.87	93.79	**90.43**	54.51	41.65	84.27	75.21
SA-Gate	85.84	80.64	93.65	88.51	**86.46**	72.71	85.70	**72.89**	92.18	89.39	**62.70**	40.59	**84.42**	74.12
TSNet	85.22	76.85	91.85	86.65	**88.52**	67.98	78.75	67.49	78.22	76.85	37.49	30.85	76.68	67.78
QGN	91.33	81.91	90.68	86.19	87.73	73.47	83.25	72.51	94.28	**91.47**	53.59	39.11	83.48	74.11
ESANet	91.38	**82.92**	**93.69**	89.82	87.10	73.16	82.48	70.81	93.08	88.53	55.68	43.38	83.90	74.77
Proposed	90.19	**82.83**	93.50	89.78	86.14	**74.09**	**87.93**	**74.22**	**94.38**	90.25	**58.84**	42.29	**85.16**	**75.58**

we use nDSMs, but the height of objects in remote sensing images is not uniform, leading to an equal elevation between objects, such as cars and low vegetation. Consequently, nDSMs introduce excessive noise for parsing remote sensing images.

The experimental results demonstrate that the proposed CEGFNet can remove redundant noise in cross-modal data and extract complementary features. In general, methods that use multimodal data outperform those that use only one data source. As DSMs provide elevation data of objects, multimodal methods can outperform single-modal methods for parsing objects, such as buildings and trees.

2) Qualitative Evaluation: The scene parsing results of various images from the test set are shown in Fig. 5. The figure shows comparisons with the results obtained from DeepLab v3+, which achieves the highest performance among the single-modal methods, and those obtained from all the multimodal comparison methods. The proposed CEGFNet accurately segments different objects under various challenging scenes, including those with objects shadowed by trees or buildings (first, second, fourth, sixth, and eighth rows) and cars close together (second and fourth rows). These results demonstrate that CEGFNet uses global context information and nonlocal information to eliminate the disturbance caused by large-scale variations in objects and shadows of trees or buildings appearing in remote sensing images. Moreover, the object boundaries and details (third and fifth rows) provided by CEGFNet are clearer and sharper than those provided by the comparison methods.

TABLE III
RESULTS OF ABLATION STUDIES ON CEGFNET

Methods	GFM	GCM	MAD	Vaihingen		Potsdam	
				mAcc	mIoU	mAcc	mIoU
WO/GCM		√	√	88.37	79.96	84.84	75.34
WO/GFM	√		√	86.54	78.65	84.38	75.51
WO/MAD	√	√		85.85	77.40	83.58	73.98
Proposed	√	√	√	**89.27**	**80.86**	**85.06**	**75.89**

E. Ablation Studies

We conducted various ablation studies to further investigate the contributions of each component in CEGFNet. Table III lists the results from the three evaluated components of the CEGFNet, where √ indicates that the corresponding module is retained, and no mark indicates that the corresponding module is removed. Supervision was conducted using the cross-entropy loss. We only replaced the GFM, GCM, and MAD but retained the other components unchanged in CEGFNet in each study. The results indicate that the GFM, GCM, and MAD contribute to the performance of CEGFNet. The complete architecture achieves the highest performance. If the GFM, GCM, or MAD is removed, the parsing performance drops on all indicators, confirming their effectiveness.

1) CEGFNet Without GFM (WO/GFM): We replaced the GFM with elementwise addition at the corresponding levels along the IRRG and nDSM streams. Fig. 6 shows that removing the GFM impedes complete scene parsing for low-quality nDSMs. In addition, the scene parsing maps in Fig. 6 are

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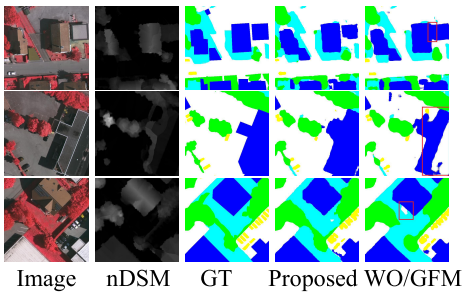


Fig. 6. Scene parsing maps obtained from CEGFNet without GFM (WO/GFM) and complete CEGFNet.

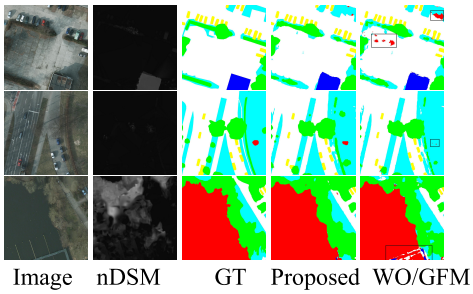


Fig. 7. Scene parsing maps obtained from CEGFNet without GCM (WO/GCM) and complete CEGFNet. Black boxes: most notable differences.

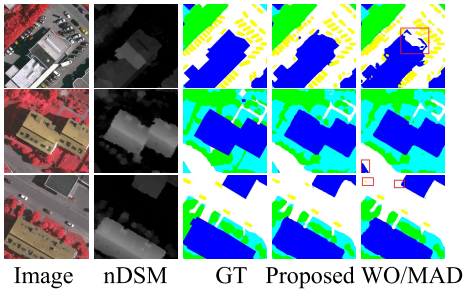


Fig. 8. Scene parsing maps obtained from CEGFNet without MAD (WO/MAD) and complete CEGFNet. Red boxes: most notable differences.

misleading because the top area of the building in the hyperspectral image is similar to that in the shaded area. Without integrating IRRG and nDSM through the GFM, part of the top of the building is incorrectly divided into other objects. These results show the effectiveness and superiority of the GFM for IRRG and nDSM fusion.

2) CEGFNet Without GCM (WO/GCM): Without the GCM, we considered that the number of channels in the output of each GFM was excessively numerous and variable. Thus, we replaced the GCM with a simple 1×1 convolutional layer to limit the sizes of the channels. Table III shows that the complete CEGFNet is notably better than its variant without the GCM. Fig. 7 shows that removing the GCM causes inadequate parsing of objects with large-scale variations. These experimental results show the effectiveness and superiority of the GCM in addressing scale variations (because the GCM increases the receptive field while maintaining the feature map resolution, and the attention mechanism reduces feature redundancy) that often appear in remote sensing images.

3) CEGFNet Without MAD (WO/MAD): For the MAD ablation study, we replaced MAD with concatenation followed by a convolution with a 3×3 kernel. Then, we used bilinear interpolation to increase the resolution of the feature map. Table III shows that CEGFNet significantly outperforms its variant without the MAD. Moreover, Fig. 8 shows that removing the MAD leads to incomplete parsing and segmentation failure for shadowed objects. One of the reasons for the failure of segmentation of the shadowed part is that the receptive field is not large enough. Although we have further expanded the receptive field in the GCM, not all input pixels can be calculated due to the discontinuity of the convolution kernel for dilated convolution; hence, we use the self-refining part in MAD. One of the features of this part is to capture the remote dependence relationship directly by calculating the interactions between any two positions instead of being limited to adjacent points, which is equivalent to constructing a convolution kernel with the same size as the feature graph to maintain more information. Therefore, we apply the MAD to use multiscale information from the GCM. As a result, during top-down decoding, the multiscale features are integrated and refined gradually. These results indicate that MAD promotes accurate scene parsing.

Overall, the proposed CEGFNet outperforms its variants in terms of all evaluation criteria, and its superior results can be confirmed visually. Therefore, each CEGFNet component contributes to the scene parsing of multimodal HSR remote sensing images.

V. Conclusion

We propose CEGFNet, the first scene parsing framework for multimodal HSR remote sensing images. The framework aims to take advantage of commonalities among features to fuse multimodal information. First, we use two common backbone branches to extract features from hyperspectral images and DSMs. Then, the GFM fuses feature maps of the corresponding sizes from the hyperspectral and DSM modalities. In addition, the GCM combines global context cues to handle large changes in object scales. Furthermore, for multilevel and multiscale feature fusion to provide the final scene parsing map, we apply the MAD, a powerful decoder. The proposed CEGFNet and SOTA scene parsing methods were evaluated on two benchmark remote sensing datasets, and the proposed CEGFNet achieved the highest performance. In addition, the effectiveness of the proposed CEGFNet to handle common problems in scene parsing was verified through several ablation studies. In future work, we will apply CEGFNet to other machine vision tasks, such as salient object detection and crowd counting.

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